Domain Understanding

Project name: “Pet Macher”

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Contents

Central Research Question 3

Exploratory Research 3

- Research Methods 3
- Research Findings and Insights (initial) 3

Analytic Approach 3

- Target Variable 3
- Type of Problem 3
- Potential Machine Learning Models 4
  - Linear Regression 4
  - Decision Trees 4
  - Random Forest Regressor 4
  - K-Nearest Neighbors (KNN) 4
- Evaluation Metrics 4

Data Requirements 5

- Objectives 5
- Characteristics 5
- Sources 5
- Legality & Ethics 5
- Version Control 5

Iterative Process 5

Conclusion 6

Central Research Question

How can we use shelter animal data and adopter preferences to predict a match percentage that increases the chance of animals finding their forever homes and adopters finding their best-matched pets?

Exploratory Research

Research Methods

To explore this question, I will:

- Review studies and resources about pet adoption, compatibility factors, and matchmaking systems.
- Collect data from animal shelters (species, age, breed, behavior, health status).
- Include adopter preferences (activity level, household type, experience with pets).
- Analyze existing adoption platforms or tools to see what works and what doesn't.
- Gather informal feedback from shelter employees and potential adopters.

Research Findings and Insights (initial)

- Shelters want to increase adoption rates but also reduce returns caused by poor matches.
- Adopters often focus on surface traits (looks or breed) rather than compatibility factors.
- A match percentage makes adoption decisions clearer and reduces uncertainty.
- Staff find that compatibility tools can save time in guiding adopters.

These insights show that the project needs to balance data-driven matchmaking with human-understandable results that both adopters and shelter staff can trust.

Analytic Approach

Target Variable

The target variable is the match percentage between an adopter and a shelter animal, expressed as a score from 0–100. This score represents how well the animal's traits align with the adopter's preferences and lifestyle.

Type of Problem

This is primarily a **regression problem**, since the outcome is a continuous percentage score (0–100).

Alternative approaches:

- **Classification:** converting percentages into categories (Low, Medium, High match).
- **Ranking:** showing the **top 5 to 10 matches** to highlight the best adoption options.

For this stage, regression provides the most detailed and flexible results.

Potential Machine Learning Models

Linear Regression

- **Why use it:** It's the simplest model and acts as a baseline. It checks if there's a roughly straight-line relationship between adopter preferences and animal traits.
- **Example:** If more time at home always increases the match percentage with high-need pets, linear regression will show that.
- **Good for:** Getting an initial sense of whether the problem can be solved with simple math before moving to complex models.

Decision Trees

- **Why use it:** Decision Trees split data into if - then rules, which makes results easy to explain.
- **Example:** "If adopter wants an active pet > check if the dog's activity level is high > if yes, increase match score."
- **Good for:** Capturing more realistic conditions, since pet matching isn't always linear. Also, very interpretable for shelters and adopters.

Random Forest Regressor

- **Why use it:** This method builds many decision trees and combines them, which makes predictions more accurate and stable. It also shows which features matter most (lifestyle may influence matching more than age).
- **Example:** It won't overfit to one weird case (like a single unusual adopter–pet combo) because it balances across many trees.
- **Good for:** Reliability and finding feature importance, so you can explain why the system matched an adopter with a pet.

K-Nearest Neighbors (KNN)

- **Why use it:** This model finds adopters similar to the current one and looks at what pets matched well for them. It works like a recommendation system.
- **Example:** If adopters with similar profiles (active lifestyle, kids at home) matched well with certain dogs, the model will suggest those as top matches.
- **Good for:** Personalization and recommendations based on similar cases.

The final choice will depend on dataset size and interpretability, with the priority being trustworthy and explainable predictions.

Evaluation Metrics

- **Mean Absolute Error (MAE):** average difference between predicted and actual match percentage.
- **R^2 Score:** to check how much variation is explained by the model.
- **Top-K Accuracy:** whether the animal the adopter eventually chooses is in the model's top 5 matches.

Data Requirements

Objectives

The project aims to:

- Help animals find their forever homes faster.
- Support adopters in making more informed decisions.
- Assist shelters in reducing failed adoptions and returns.

Success will be measured by the clarity of the match results and whether the model can realistically support shelter adoption workflows.

Characteristics

The required data will include:

- **Animal data:** species, breed, age, size, activity level, temperament, health conditions.
- **Adopter data:** household type, family/children, lifestyle (active/sedentary), experience with pets, preferences.
- **Output:** match percentage (0–100).

All data will be numerical or converted into numerical values (categorical variables encoded as integers).

Sources

- Shelter employee inputs.
- Public adoption platforms (for reference datasets).
- Internet resources and statistics

Legality & Ethics

Since this project involves personal preferences and animal data, ethical considerations include:

- Ensuring adopter data (household details) is handled responsibly and with consent.
- Avoiding bias in recommendations (favoring certain breeds unfairly).
- Keeping the model transparent so adopters understand why a pet is recommended.

Version Control

Version control will be maintained via a Git repository. Since some data may be sensitive (stakeholder inputs, adoption preferences), the repository will remain private.

Iterative Process

- **Proposal (Week 1):** Define project scope and research.
- **Provisioning (Weeks 2–3):** Collect and prepare data, define features.
- **Model Development (Weeks 4–5):** Train and test different ML models.
- **Evaluation & Delivery (Week 6):** Validate results, document findings, and finalize.

Conclusion

This document outlines the approach for Pet Matcher, an AI-powered system to predict compatibility between shelter animals and adopters. By combining animal traits, adopter preferences, and machine learning, the model aims to provide a percentage-based match score that is both accurate and easy to interpret.

The project emphasizes not just predicting a match, but also building trust and transparency so both adopters and shelters can use it confidently. This foundation sets the direction for data collection, modeling, and evaluation over the planned six-week timeline.